

Strategic coexistence theory for evolutionary games

Sang Woo Park^{1,2,*}

1 School of Biological Sciences, Seoul National University, Seoul, Korea

2 Institute for Data Innovation in Science, Seoul National University, Seoul, Korea

*Corresponding author: sangwoopark@snu.ac.kr

Abstract

Evolutionary game theory and ecological coexistence theory both seek to predict the outcome of competition between biological entities, be they strategies or species, but the two fields have relied on largely separate approaches. Replicator equations provide a foundation for predicting the outcome of strategic competition, yet they do not explicitly decompose the mechanisms underlying these outcomes. Inspired by modern coexistence theory from community ecology, we develop Strategic Coexistence Theory (SCT), which translates payoff-based invasion conditions into strategic niche differences and fitness differences between any two competing strategies described by classic replicator dynamics. SCT recovers the classic classification of two-strategy games, distinguishing competitive exclusion, coexistence, and priority effects within a shared niche–fitness difference space. Applying SCT to five mechanisms for the evolution of cooperation further reveals that these mechanisms promote cooperation through distinct dynamical routes: kin selection, network reciprocity, and group selection primarily reduce fitness differences, whereas direct and indirect reciprocity destabilize competition and generate priority effects. Finally, applying SCT to microbial public-goods competition shows that nonlinear microbial growth can both stabilize and equalize competition between cooperators and defectors, allowing coexistence. Together, these results show that SCT provides a common language for comparing coexistence mechanisms across evolutionary games.

Significance statement

Evolutionary game theory (EGT) provides a powerful framework for predicting the outcome of competition between strategies, including the dominance of one strategy, coexistence of multiple strategies, and bistability. However, a framework for quantifying the mechanisms that allow strategies to coexist remains elusive, even though stable polymorphisms, such as the coexistence of cooperators and defectors, are widely observed across biological systems. To address this gap, we developed Strategic Coexistence Theory (SCT), which extends Modern Coexistence Theory from community ecology to evolutionary games. Applying SCT to classic games, mechanisms of cooperation, and microbial public-goods competition shows that seemingly

38 similar evolutionary outcomes can arise through distinct mechanisms. This approach
39 links ecological theory with replicator dynamics, providing a complementary tool for
40 analyzing evolutionary games.

41 Introduction

42 Understanding the mechanisms that promote the evolutionary stability of competing
43 strategies is a fundamental aim in evolutionary biology (Lewontin, 1961; Slobodkin,
44 1964; Smith and Price, 1973; Taylor and Jonker, 1978). To this end, evolutionary
45 game theory (EGT) and replicator dynamics have provided key foundations, allow-
46 ing us to predict the outcome of strategic competition through payoff inequalities,
47 invasion criteria, and stability analysis (Schuster and Sigmund, 1983; Hofbauer and
48 Sigmund, 1998; Cressman, 2003; Cressman and Tao, 2014). Recently, there has been
49 growing interest in characterizing similarities between replicator dynamics and the
50 dynamics of competing species (Gokhale and Traulsen, 2025; Tarnita and Traulsen,
51 2025; Allesina, 2026), building on the classical mathematical equivalence between
52 replicator equations and generalized Lotka–Volterra models (Hofbauer, 1981; Page
53 and Nowak, 2002). In particular, Tarnita and Traulsen (2025) recently argued that a
54 closer synthesis between ecology and evolutionary game theory can provide new in-
55 sights into ecological and evolutionary assembly, competition, and coexistence. This
56 renewed perspective between ecological and evolutionary dynamics suggests that evo-
57 lutionary games may be interpreted not only as payoff-based competition but also
58 as problems of strategic coexistence.

59 Interestingly, even a simple two-strategy game can yield heterogeneous outcomes
60 that mirror outcomes in community ecology (Hauert, 2002; Szabó and Fath, 2007;
61 Cooney, 2020; Tarnita and Traulsen, 2025): the dominance of a single strategy (pris-
62 oner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability
63 (stag-hunt games). Such differences in conditions that promote evolutionary stability
64 and dominance of a single strategy and those that permit strategy coexistence indi-
65 cate that EGT can be understood from the perspective of coexistence theory (Motro,
66 1991; Doebeli et al., 2004; Hauert and Doebeli, 2004; Doebeli and Hauert, 2005; Gore
67 et al., 2009; Souza et al., 2009; Archetti and Scheuring, 2011; Steidinger and Bever,
68 2014; Tarnita and Traulsen, 2025). In fact, stable polymorphisms of evolutionary
69 strategies, such as coexistence of cooperators and defectors, are observed in micro-
70 bial and other biological systems (Gore et al., 2009; Cordero et al., 2012; Kocher
71 et al., 2018; Pollak et al., 2021; Lin et al., 2024), but coexistence-theoretic perspec-
72 tive has often remained implicit. Recasting game dynamics as a coexistence problem
73 immediately raises the following question: when two strategies coexist, does the
74 corresponding mechanism stabilize or equalize competition (Chesson, 2000)? While
75 EGT allows us to predict whether two strategies can coexist or not, a framework for
76 quantifying coexistence mechanisms between competing strategies remains elusive.

77 Modern coexistence theory (MCT) from community ecology presents a particu-

larly useful framework for quantifying coexistence mechanisms (Chesson, 2000; Adler et al., 2007; Godoy et al., 2014; Godoy and Levine, 2014; Kraft et al., 2015; Adler et al., 2026). MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to overcome average fitness differences (Chesson, 2000; Adler et al., 2007; Kraft et al., 2015). In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor (Chesson, 2000). In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions (Taylor and Jonker, 1978; Cressman and Tao, 2014; Tarnita and Traulsen, 2025). Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each entity to compete under the limiting environments created by its competitor (Chesson, 2000). This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another (Taylor and Jonker, 1978; Cressman and Tao, 2014; Tarnita and Traulsen, 2025).

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains (Sieben et al., 2022; Park et al., 2024, 2026). Inspired by this effort, we present Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two-strategy games, providing a reinterpretation of classic results from coexistence-theoretic perspective. We then apply SCT to comparing five mechanisms for the evolution of cooperation, revealing that direct and indirect reciprocity destabilize competition between cooperators and defectors, thus pushing the system towards priority effect regime. Finally, we consider a case study of microbial cooperation, where nonlinear microbial growth can both stabilize and equalize competition between cooperation and defection, allowing coexistence of strategies. Together, these examples show that SCT provides a common, ecological language for comparing the mechanisms underlying strategic evolution.

Strategic coexistence theory

Here, we present a brief derivation of strategic coexistence theory (SCT) (Figure 1). To do so, we begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs (Schuster and Sigmund, 1983; Hofbauer and Sigmund, 1998; Cressman, 2003; Cressman and Tao, 2014):

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j .

Because payoffs in many games can be negative, payoff ratios can be difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

This exponential transformation is convenient because it preserves the payoff ordering and does not alter the invasion conditions. Therefore, this allows us to compare strategic environments through ratios, as in modern coexistence theory. For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

When $1 - \rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition. Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

In other words, strategy j has an average competitive advantage over strategy i ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated by itself and its competitor.

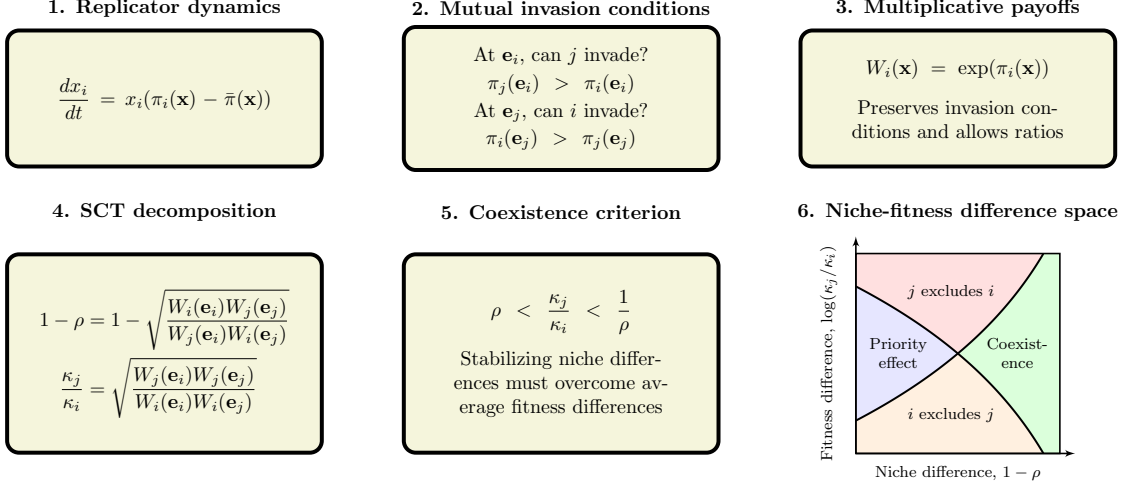


Figure 1: **A schematic diagram summarizing strategic coexistence theory.** Strategic coexistence theory decomposes pairwise replicator dynamics into stabilizing and equalizing mechanisms. Mutual invasion conditions can be rewritten on a multiplicative payoff scale, allowing us to derive strategic niche difference $1 - \rho$ and fitness difference κ_j/κ_i based on multiplicative payoff ratios. Then, the inequality $\rho < \kappa_j/\kappa_i < 1/\rho$ determines mutual invasion.

149 Together, for any two strategies described by the classic replicator equation, mu-
 150 tual invasion requires stabilizing niche differences to overcome average fitness differ-
 151 ences. Specifically, the condition for mutual invasion can be rewritten as (Chesson,
 152 2000):

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

153 This inequality applies to any games described by the classic replicator equation
 154 because ρ and κ_j/κ_i can be calculated directly from ratios of multiplicative pay-
 155 offs. This decomposition not only allows us to understand how different mechanisms
 156 contribute to the coexistence or exclusion of competing strategies but also provides
 157 shared axes for comparing different games. Detailed derivations are provided in
 158 Supplementary Text S1.

159 We note that the absolute magnitudes of niche and fitness differences are sensitive
 160 to payoff scaling because they are defined on an exponentiated payoff scale. However,
 161 the qualitative outcome of strategic competition under SCT is preserved because the
 162 exponential transformation does not alter invasion conditions.

163 Classic two-strategy games

164 As an illustrative example, we begin with classic two-strategy games (Hauert, 2002;
 165 Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025), which can be

described by the following payoff matrix (Figure 2A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

Here, the first and second rows represent strategies 1 and 2, respectively, and the columns represent the strategy of the interacting opponent. Writing x_1 to denote the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

The signs of $T - R$ and $S - P$ determine whether each strategy can invade the other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition the game space into four regimes (Figure 2B): prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt game ($R > T, P > S$), and harmony game ($R > T, S > P$). In standard EGT, these outcomes are obtained by analyzing the stability of equilibrium conditions. For simple games, this analysis is straightforward.

A coexistence-theoretic approach recovers the same classification but provides a different interpretation (Figure 2C). For a two-strategy game determined by this payoff matrix, the niche difference and fitness difference are given by the following set of equations (Supplementary Text S2):

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to $T > R$ and $S > P$. This corresponds to the hawk-dove game, in which each strategy can invade its competitor when rare and the two strategies coexist. The remaining games also have natural interpretations in niche-fitness difference space. Prisoner's dilemma and harmony games represent competitive exclusion regimes, where one strategy has a fitness advantage that exceeds niche differences. In contrast, stag-hunt games represent a priority-effect regime, where neither strategy can invade the other and the outcome depends on initial conditions. Therefore, coexistence theory not only allows us to recover the classic results from two-strategy games but also provides new interpretations for these results in terms of niche and fitness differences. Building on this, we next apply this framework to comparing core mechanisms of cooperation.

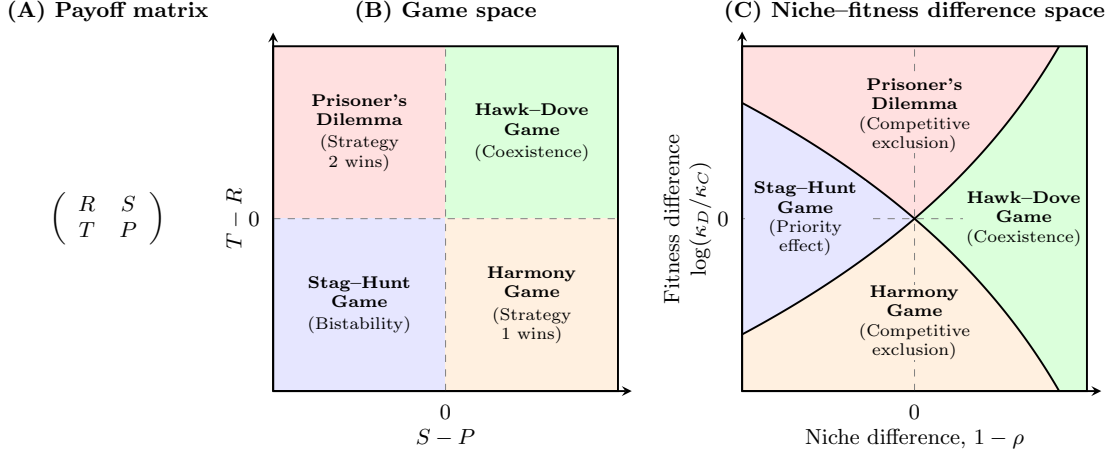


Figure 2: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk-dove games allow coexistence; and stag-hunt games result in bistability. (C) Mapping of the same four games onto niche-fitness difference space. Hawk-dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag-hunt games correspond to the priority-effect regime. $R > P$ is assumed throughout, but this does not affect coexistence outcome in any way.

Comparisons of mechanisms for the evolution of co-operation

Even though cooperative behavior can be found across a wide range of biological systems, a simple prisoner's dilemma shows that cooperation is not an evolutionarily stable strategy (Axelrod and Hamilton, 1981; May, 1987). Thus, understanding mechanisms that promote the evolution of cooperation has remained a central goal in evolutionary biology (West et al., 2007; Fletcher and Doebeli, 2008; Clutton-Brock, 2009; Kingma et al., 2014; Rakoff-Nahoum et al., 2016; Hilbe et al., 2018; Apicella and Silk, 2019; West et al., 2021; Lee et al., 2022; Palmer and Foster, 2022; Michel-Mata et al., 2024). Here, we compare five classic mechanisms for the evolution of cooperation using SCT (Nowak, 2006) and ask how they differ from the perspective of SCT.

Specifically, we consider the following mechanisms: kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection (Figure 3A). Kin selection favors cooperation when two interacting individuals are genetic rela-

209 tives, where the genetic relatedness r must be greater than the cost-to-benefit ratio:
 210 $r > c/b$ (Hamilton, 1964). Direct reciprocity favors cooperation when current co-
 211 operation may promote future cooperation by the same individual through repeated
 212 encounters (Trivers, 1971; Axelrod and Hamilton, 1981). In this case, the probability
 213 of repeated interaction (i.e., “next round”) must be greater than the cost-to-benefit
 214 ratio: $w > c/b$. Indirect reciprocity favors cooperation when current cooperation
 215 may promote future cooperation by different individuals through reputation (Nowak
 216 and Sigmund, 1998; Wedekind and Milinski, 2000; Nowak and Sigmund, 2005). Thus,
 217 the social acquaintanceship (i.e., the probability of knowing someone’s reputation)
 218 must be greater than the cost-to-benefit ratio: $q > c/b$. Network reciprocity favors
 219 cooperation via clustering of cooperators, requiring the benefit-to-cost ratio to be
 220 greater than the average number of neighbors: $b/c > k$ (Nowak and May, 1992;
 221 Lieberman et al., 2005). Finally, group selection favors cooperation via multilevel
 222 selection, where groups with more cooperators exhibit faster growth at the group
 223 level even if defectors reproduce faster at the individual level (Traulsen and Nowak,
 224 2006). This requires $b/c > 1 + (n/m)$, where n is the maximum group size and
 225 m is the number of groups. All of these mechanisms can be described by a simple
 226 2×2 payoff matrix (Nowak, 2006), from which the corresponding niche and fitness
 227 differences can be computed using Eq. (10) and Eq. (11). Details on niche and fitness
 228 difference calculations are presented in Supplementary Text S3.

229 First, the prisoner’s dilemma provides a baseline scenario, where defection is
 230 evolutionarily stable (Figure 3A). From the perspective of SCT, there is complete
 231 niche overlap, and therefore defection competitively excludes cooperation (Figure
 232 3B). Building on this, SCT shows that five mechanisms promote evolutionary sta-
 233 bility of cooperation via different dynamical routes (Figure 3B). For example, kin
 234 selection allows the fitness of cooperation to increase as genetic relatedness r increases
 235 (Figure 3B). However, it does not affect the niche difference between cooperation and
 236 defection. Similarly, network reciprocity and group selection modulate the fitness of
 237 cooperation without causing any changes in strategic niches. In contrast, direct
 238 and indirect reciprocity destabilize competition while increasing the fitness of coop-
 239 eration, thereby pushing the competition outcome towards a priority effect regime.
 240 Equivalently, these two mechanisms promote bistability of defection and cooperation.

241 SCT further reveals another difference between direct and indirect reciprocity.
 242 In particular, direct reciprocity is unable to generate sufficient fitness difference to
 243 competitively exclude defection even when the probability of next round w reaches
 244 1 (Supplementary Text S3). In contrast, indirect reciprocity is able to generate
 245 sufficient fitness difference to push the system towards the boundary between priority-
 246 effect regime and competitive-exclusion regime as social acquaintanceship q reaches 1,
 247 indicating that an additional mechanism may be able to promote complete dominance
 248 of cooperation (Supplementary Text S3).

249 We note that none of these mechanisms allow coexistence of cooperation and de-
 250 defection. In many biological systems, such coexistence is often observed (Gore et al.,
 251 2009; Cordero et al., 2012; Kocher et al., 2018; Pollak et al., 2021; Lin et al., 2024).

(A)

Prisoner's dilemma	Kin selection
$\begin{pmatrix} b-c & -c \\ b & 0 \end{pmatrix}$	$\begin{pmatrix} (b-c)(1+r) & br-c \\ b-r & 0 \end{pmatrix}$
Direct reciprocity	Indirect reciprocity
$\begin{pmatrix} (b-c)/(1-w) & -c \\ b & 0 \end{pmatrix}$	$\begin{pmatrix} b-c & -c(1-q) \\ b(1-q) & 0 \end{pmatrix}$
Network reciprocity	Group selection
$\begin{pmatrix} b-c & H-c \\ b-H & 0 \end{pmatrix}$	$\begin{pmatrix} (b-c)(m+n) & (b-c)m-cn \\ bn & 0 \end{pmatrix}$

(B)

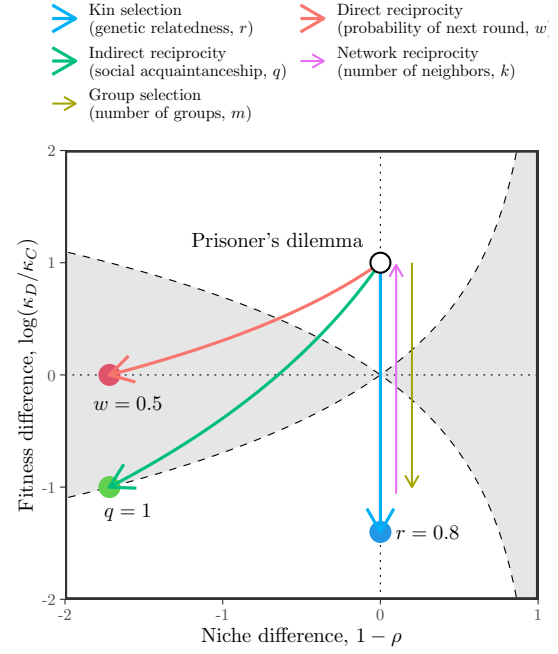


Figure 3: **Comparisons of five mechanisms for the evolution of cooperation using SCT.** (A) Payoff matrices for the baseline scenario (prisoner's dilemma) and five mechanisms for the evolution of cooperation (Nowak, 2006). Prisoner's dilemma is characterized by the benefit for the recipient b and the cost for the donor c . Network reciprocity is parameterized based on $H = [(b-c)k - 2c]/[(k+1)(k-2)]$. Parameters for all other mechanisms are described in panel B. (B) Changes in niche and fitness differences driven by each mechanism. Each arrow represents the direction of change as we increase the corresponding parameter, indicated in figure legends. Note that kin selection, network reciprocity, and group selection all result in changes in fitness difference without causing any changes in niche difference. Therefore, we indicate the direction of change for network reciprocity and group selection separately to avoid overlapping arrows. For simplicity, we focus on the effect of number of groups m for group selection; increasing the number of maximum group size n has the opposite effect (Supplementary Text S3).

Therefore, we move on to the final example, applying SCT to microbial cooperation, where nonlinear benefits and environmental context can promote coexistence of cooperators and defectors.

Microbial cooperation and public goods

Here, we consider a simple, phenomenological model of competition between cooperating and cheating yeast in a cooperative sucrose-metabolism system, where invertase

258 production by cooperators allows cheaters to utilize glucose, a public good released
 259 through sucrose hydrolysis (Carlson and Botstein, 1982; Dickinson, 1998; Gore et al.,
 260 2009). Specifically, invertase production by cooperators incurs a cost c and gener-
 261 ates a total benefit of 1 that is captured privately with efficiency ϵ . Assuming a
 262 linear relationship between glucose availability and microbial growth, the payoffs for
 263 cooperators π_C and cheaters (defectors) π_D can then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

264 where the use of glucose depends on the fraction of cooperators f . The dynamics of
 265 cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

266 For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of
 267 f (Figure 4A). Therefore, cooperation is favored when efficiency is greater than the
 268 cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than
 269 efficiency, $c > \epsilon$ (Gore et al., 2009).

270 Our framework allows us to reinterpret this result from the perspective of coexis-
 271 tence theory. In particular, the two strategies have no niche difference, corresponding
 272 to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by
 273 the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

274 Therefore, this model always predicts competitive exclusion, except in the neutral
 275 case $c = \epsilon$. The outcome is determined entirely by the fitness difference between
 276 cooperators and defectors, rather than by stabilizing niche differences. This example
 277 resembles the prisoner's dilemma discussed in the previous example (Figure 3).

278 In real biological systems, the effect of glucose on microbial growth follows a
 279 nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

280 where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial
 281 growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are
 282 described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\{\epsilon + f(1 - \epsilon)\}^\alpha - c - \{f(1 - \epsilon)\}^\alpha]. \quad (19)$$

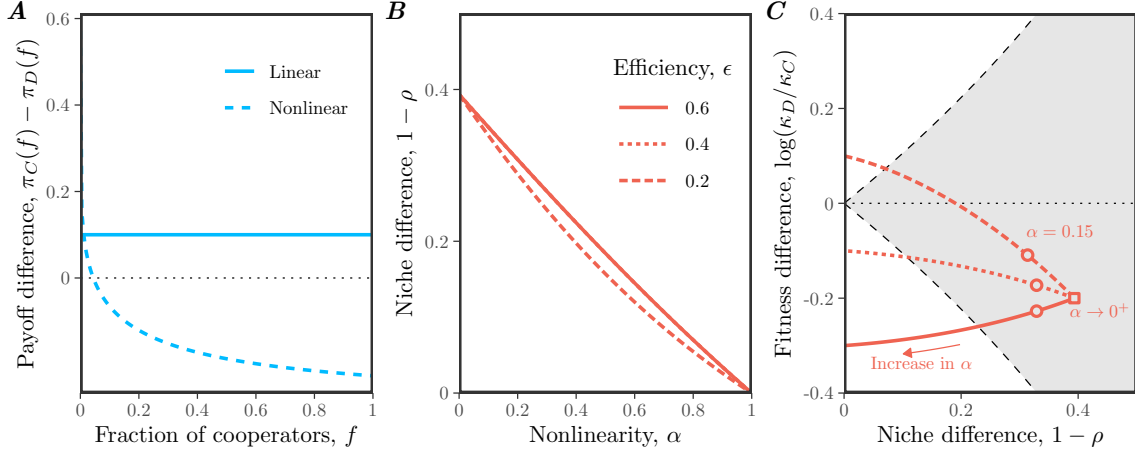


Figure 4: **Effects of nonlinear microbial growth on coexistence of cooperation and defection.** (A) Payoff differences between cooperation and defection for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.3$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . Note that lines for $\epsilon = 0.6$ and $\epsilon = 0.4$ are nearly overlapping and are indistinguishable. (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha \rightarrow 0^+$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.3$, while allowing ϵ and α to vary.

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction f , which can promote the coexistence of cooperators and defectors (Figure 4A): cooperators have a competitive advantage when rare, whereas defectors have a competitive advantage when cooperators are abundant.

From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient stabilizing niche difference to overcome the fitness difference. In particular, the niche and fitness differences for this model correspond to:

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For example, the cost of cooperation does not affect niche difference and only affects fitness difference. Instead, niche difference is generated by the concave relationship between glucose availability and microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0 (Figure 4B,C). In contrast, as $\alpha \rightarrow 0^+$, the fitness

295 difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

296 which no longer depends on the private capture efficiency ϵ (Figure 4C). Thus, strong
297 concavity not only generates stabilizing niche differences, but also acts as an equal-
298 izing mechanism (Figure 4C). Such stabilizing mechanisms are key ingredients for
299 coexistence of cooperators and defectors.

300 Discussion

301 Standard evolutionary game theory and replicator equations provide powerful tools
302 for predicting the outcome of strategy competition, including strategic dominance,
303 coexistence, and bistability (Motro, 1991; Doebeli et al., 2004; Hauert and Doebeli,
304 2004; Doebeli and Hauert, 2005; Gore et al., 2009; Souza et al., 2009; Archetti and
305 Scheuring, 2011; Steidinger and Bever, 2014; Tarnita and Traulsen, 2025). However,
306 these outcomes are typically understood by analyzing payoff differences within in-
307 dividual games, which makes it difficult to compare coexistence mechanisms across
308 models. Building on Chesson’s Modern Coexistence Theory from community ecol-
309 ogy (Chesson, 2000), we developed a strategic coexistence theory (SCT) that allows
310 us to decompose strategy competition in terms of stabilizing niche differences and
311 fitness differences. This decomposition provides a unifying framework for comparing
312 different games from the perspective of coexistence theory, allowing us to identify
313 stabilizing and equalizing mechanisms for strategy competition.

314 To demonstrate the utility of SCT, we considered three case studies. First, the
315 analysis of classic two-strategy games showed that SCT recovers the original clas-
316 sification for these classic games, highlighting the similarities between evolutionary
317 games and community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020;
318 Tarnita and Traulsen, 2025). Then, we compared different mechanisms for the evo-
319 lution of cooperation, which showed that each mechanism promotes cooperation via
320 different dynamical routes (Nowak, 2006). Kin selection, network reciprocity, and
321 group selection primarily affect the fitness difference between cooperation and defec-
322 tion, whereas direct and indirect reciprocity can destabilize competition and generate
323 priority effects. Finally, our analysis of microbial cooperation showed that coexis-
324 tence between cooperators and defectors requires more than reducing the fitness cost
325 of cooperation: nonlinear benefits can generate stabilizing niche differences while
326 also reducing fitness asymmetry (Gore et al., 2009). Taken together, these examples
327 show that SCT can provide new insights into ecological underpinnings of strategy
328 competition.

329 There are several limitations to SCT. First, similar to MCT from community
330 ecology, SCT only allows for comparisons of pairwise strategies and therefore cannot
331 determine the outcomes of games with more than two strategies (Chesson, 2000).
332 Second, we focused on classic replicator equations, but Tarnita and Traulsen (2025)
333 recently pointed out that these classic formulations implicitly ignore differences in

intrinsic growth rates. In such cases, replicator equations may fail to predict the outcome of competition (Tarnita and Traulsen, 2025), and a full Lotka-Volterra model is needed to describe the abundances, rather than frequencies, of strategies. Thus, SCT inherits the same assumptions as the classic replicator framework: it is most appropriate when strategy competition can be described in terms of relative frequencies alone (Tarnita and Traulsen, 2025). When strategies differ intrinsically in growth, MCT may be required instead to tease apart mechanisms of strategy coexistence. Finally, we have focused on simple examples to illustrate the utility of SCT, but many games involve additional complexities, including population structure (Nowak et al., 2010), stochasticity (Wang et al., 2025; Bodin et al., 2026), and explicit ecological feedback (Tarnita and Traulsen, 2025). Extending SCT to these settings will be necessary to determine how broadly stabilizing and equalizing mechanisms can be compared across evolutionary games.

In conclusion, SCT provides a bridge between evolutionary game theory and community ecology, allowing strategy competition to be understood from the perspective of coexistence theory. Therefore, rather than replacing existing EGT or replicator-dynamics frameworks, SCT provides a complementary tool for teasing apart the coexistence mechanisms underlying evolutionary games. More broadly, our study builds on recent efforts to extend MCT beyond species competition (Park et al., 2024, 2026), providing a foundation for building unifying coexistence theory for competing entities, including species, pathogen strains, and behavioral strategies.

Acknowledgement

We thank Jonathan Levine for the helpful discussion. S.W.P. was supported by the New Faculty Startup Fund from Seoul National University, the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2026-25474574), and the Global-LAMP Program of the National Research Foundation of Korea (NRF) grant funded by the Ministry of Education (No. RS-2023-00301976).

Competing interests

The authors declare no competing interest.

Data availability

All data and code are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/coexistence-strategy>).

367 Supplementary Text

368 S1 Derivation of the strategic coexistence theory

369 To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

370 where x_i represents the frequency of strategy i , π_i represents the payoff of strategy
 371 i , and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two
 372 competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

373 These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) =$
 374 $\exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

375 Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

376 By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

377 Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

378 we obtain the following inequality that defines conditions for mutual invasion and
 379 long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

380 S2 Applications to classic two-strategy games

381 Consider a two-strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S11})$$

382 There are four classic games that we can define from this simple payoff matrix:
 383 prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt
 384 game ($R > T, P > S$), and harmony game ($R > T, S > P$). Here, we show that
 385 each game corresponds to a distinct region in the niche-fitness difference space under
 386 SCT.

387 To do so, we first begin by deriving expressions for niche overlap ρ and fitness
 388 difference κ_j/κ_i . Writing x_1 to denote the frequency of strategy 1, the expected
 389 payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S12})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S13})$$

390 Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S14})$$

$$W_1(0) = \exp(S) \quad (\text{S15})$$

$$W_2(1) = \exp(T) \quad (\text{S16})$$

$$W_2(0) = \exp(P) \quad (\text{S17})$$

391 Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \quad (\text{S18})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (\text{S19})$$

392 As before, SCT allows us to predict the mutual invasion of two strategy using the
 393 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S20})$$

394 Then, simple algebra reveals that each condition in mutual invasion inequality
 395 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S21})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S22})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S23})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S24})$$

396 These inequalities then allow us to classify outcomes of two-strategy games from the
 397 perspective of SCT. Note that the inequality between R and P does not affect the
 398 outcome.

399 **S3 Applications to mechanisms for the evolution of coopera-** 400 **tion**

401 We consider five mechanisms for the evolution of cooperation, all of which can be
 402 represented as a simple 2×2 payoff matrix (Nowak, 2006). Here, we go through each
 403 mechanism and derive measures for niche and fitness differences.

404 **Prisoner's dilemma.** Here, we consider a special case of Prisoner's dilemma
 405 which can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c \\ b & 0 \end{pmatrix}, \quad (\text{S25})$$

406 where b represents the benefit for the recipient and c represents cost for the donor.
 407 In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0 \quad (\text{S26})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c). \quad (\text{S27})$$

408 Therefore, as long as the cost is greater than zero, prisoner's dilemma always results
 409 in the competitive exclusion of cooperation.

410 **Kin selection.** Kin selection can be described by the following payoff matrix:

$$\begin{pmatrix} (b - c)(1 + r) & br - c \\ b - rc & 0 \end{pmatrix}, \quad (\text{S28})$$

411 where r represents the genetic relatedness. In this case, the niche and fitness differ-
 412 ences correspond to:

$$1 - \rho = 0 \quad (\text{S29})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c - br). \quad (\text{S30})$$

413 Therefore, defection will exclude cooperation when $c > br$. Equivalently, cooperation
 414 will be favored when $r > c/b$, which recovers the classic result.

415 **Direct reciprocity.** Direct reciprocity can be described by the following payoff
 416 matrix:

$$\begin{pmatrix} (b - c)/(1 - w) & -c \\ b & 0 \end{pmatrix}, \quad (\text{S31})$$

where w represents the probability of next round. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)/(1 - w) + c - b}{2}\right) \quad (\text{S32})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{b + c - (b - c)/(1 - w)}{2}\right). \quad (\text{S33})$$

Therefore, defection will exclude cooperation when

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S34})$$

$$\iff \exp\left(\frac{(b - c)/(1 - w) + c - b}{2}\right) < \exp\left(\frac{b + c - (b - c)/(1 - w)}{2}\right) \quad (\text{S35})$$

$$\iff bw < c \quad (\text{S36})$$

However, note that the following equality holds whenever $c > 0$:

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}. \quad (\text{S37})$$

Therefore, when $w > c/b$, we will have

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1} < \rho, \quad (\text{S38})$$

which implies priority effect (equivalently, bistability). Even as $w \rightarrow 1^-$, this inequality holds and therefore, cooperation cannot competitively exclude defection.

Indirect reciprocity. Indirect reciprocity can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c(1 - q) \\ b(1 - q) & 0 \end{pmatrix}, \quad (\text{S39})$$

where q represents social acquaintanceship. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)q}{2}\right), \quad (\text{S40})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{2c - (b + c)q}{2}\right). \quad (\text{S41})$$

In this case, the condition for the dominance of defection (and therefore competitive exclusion of cooperation) can be written as:

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S42})$$

$$\iff bq < c \quad (\text{S43})$$

Moreover, the following inequality always holds as long as $c > 0$.

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}, \quad (\text{S44})$$

which implies that coexistence between cooperation and defection is not possible. Therefore, when $q < c/b$, then defection will competitively exclude cooperation. Otherwise, the system will enter the priority-effect regime.

Interestingly, when $q \rightarrow 1$, we have

$$\rho = \exp\left(\frac{b-c}{2}\right) \quad (\text{S45})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{c-b}{2}\right) \quad (\text{S46})$$

$$\frac{1}{\rho} = \exp\left(\frac{c-b}{2}\right) \quad (\text{S47})$$

$$(\text{S48})$$

In other words, the following inequality will hold:

$$\frac{1}{\rho} = \frac{\kappa_2}{\kappa_1} < \rho \quad (\text{S49})$$

Network reciprocity. Network reciprocity can be described by the following payoff matrix:

$$\begin{pmatrix} b-c & H-c \\ b-H & 0 \end{pmatrix}, \quad (\text{S50})$$

where $H = [(b-c)k - 2c]/[(k+1)(k-2)]$ and k represents the number of neighbors. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S51})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(-H + c), \quad (\text{S52})$$

which implies that there is complete niche overlap and that fitness difference decreases with H . Since H is a monotonically decreasing function of k when $k > 2$ and $b/c \geq 2$, fitness difference will increase with k . In particular, cooperation will competitively exclude defection when $H > c$, meaning that:

$$b/c > k. \quad (\text{S53})$$

Group selection. Group selection can be described by the following payoff matrix:

$$\begin{pmatrix} (b-c)(m+n) & (b-c)m - cn \\ bn & 0 \end{pmatrix}, \quad (\text{S54})$$

where m represents the number of groups and n represents the group size. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S55})$$

$$\frac{\kappa_2}{\kappa_1} = \exp((c - b)m + cn), \quad (\text{S56})$$

which implies that there is complete niche overlap. The fitness difference will increase with n but decrease with m , provided that $b > c$. In particular, cooperation will competitively exclude defection under the following condition:

$$1 + \frac{n}{m} < \frac{b}{c}. \quad (\text{S57})$$

References

- Adler, P. B., M. Detto, S. P. Ellner, T. L. Gibbs, Z. J. Gold, S. J. Leonard, J. I. Levine, A. E. Schiffer, C. Song, M. Stemkovski, et al. (2026). What have we learned from empirical applications of modern coexistence theory? *Ecology letters* 29(6), e70404.
- Adler, P. B., J. HilleRisLambers, and J. M. Levine (2007). A niche for neutrality. *Ecology letters* 10(2), 95–104.
- Allesina, S. (2026). Global stability of ecological and evolutionary dynamics via equivalence. *Proceedings of the National Academy of Sciences* 123(13), e2534915123.
- Apicella, C. L. and J. B. Silk (2019). The evolution of human cooperation. *Current Biology* 29(11), R447–R450.
- Archetti, M. and I. Scheuring (2011). Coexistence of cooperation and defection in public goods games. *Evolution* 65(4), 1140–1148.
- Axelrod, R. and W. D. Hamilton (1981). The evolution of cooperation. *science* 211(4489), 1390–1396.
- Bodin, F., G. Wang, and J. B. Plotkin (2026). Eco-evolutionary games in noisy environments. *bioRxiv*, 2026–05.
- Carlson, M. and D. Botstein (1982). Two differentially regulated mRNAs with different 5' ends encode secreted and intracellular forms of yeast invertase. *Cell* 28(1), 145–154.
- Chesson, P. (2000). Mechanisms of maintenance of species diversity. *Annual review of Ecology and Systematics* 31(1), 343–366.

474 Clutton-Brock, T. (2009). Cooperation between non-kin in animal societies. *Nature* 462(7269), 51–57.
475

476 Cooney, D. B. (2020). Analysis of multilevel replicator dynamics for general two-
477 strategy social dilemma. *Bulletin of Mathematical Biology* 82(6), 66.

478 Cordero, O. X., L.-A. Ventouras, E. F. DeLong, and M. F. Polz (2012). Public good
479 dynamics drive evolution of iron acquisition strategies in natural bacterioplankton
480 populations. *Proceedings of the National Academy of Sciences* 109(49), 20059–
481 20064.

482 Cressman, R. (2003). *Evolutionary dynamics and extensive form games*, Volume 5.
483 MIT Press.

484 Cressman, R. and Y. Tao (2014). The replicator equation and other game dynamics.
485 *Proceedings of the National Academy of Sciences* 111(supplement_3), 10810–10817.

486 Dickinson, J. R. (1998). *Metabolism and molecular physiology of Saccharomyces*
487 *cerevisiae*. Taylor & Francis.

488 Doebeli, M. and C. Hauert (2005). Models of cooperation based on the Prisoner’s
489 Dilemma and the Snowdrift game. *Ecology letters* 8(7), 748–766.

490 Doebeli, M., C. Hauert, and T. Killingback (2004). The evolutionary origin of coop-
491 erators and defectors. *science* 306(5697), 859–862.

492 Fletcher, J. A. and M. Doebeli (2008). A simple and general explanation for the evolu-
493 tion of altruism. *Proceedings of the Royal Society B: Biological Sciences* 276(1654),
494 13.

495 Godoy, O., N. J. Kraft, and J. M. Levine (2014). Phylogenetic relatedness and the
496 determinants of competitive outcomes. *Ecology letters* 17(7), 836–844.

497 Godoy, O. and J. M. Levine (2014). Phenology effects on invasion success: insights
498 from coupling field experiments to coexistence theory. *Ecology* 95(3), 726–736.

499 Gokhale, C. S. and A. Traulsen (2025). Higher-order equivalence of Lotka-Volterra
500 and replicator dynamics. *bioRxiv*, 2025–03.

501 Gore, J., H. Youk, and A. Van Oudenaarden (2009). Snowdrift game dynamics and
502 facultative cheating in yeast. *Nature* 459(7244), 253–256.

503 Hamilton, W. D. (1964). The genetical evolution of social behaviour. II. *Journal of*
504 *theoretical biology* 7(1), 17–52.

505 Hauert, C. (2002). Effects of space in 2×2 games. *International Journal of Bifur-*
506 *cation and Chaos* 12(07), 1531–1548.

507 Hauert, C. and M. Doebeli (2004). Spatial structure often inhibits the evolution of
508 cooperation in the snowdrift game. *Nature* 428(6983), 643–646.

509 Hilbe, C., Š. Šimsa, K. Chatterjee, and M. A. Nowak (2018). Evolution of cooperation
510 in stochastic games. *Nature* 559(7713), 246–249.

511 Hofbauer, J. (1981). On the occurrence of limit cycles in the volterra-lotka equation.
512 *Nonlinear Analysis: Theory, Methods & Applications* 5(9), 1003–1007.

513 Hofbauer, J. and K. Sigmund (1998). *Evolutionary games and population dynamics*.
514 Cambridge university press.

515 Kingma, S. A., P. Santema, M. Taborsky, and J. Komdeur (2014). Group augmen-
516 tation and the evolution of cooperation. *Trends in ecology & evolution* 29(8),
517 476–484.

518 Kocher, S. D., R. Mallarino, B. E. Rubin, D. W. Yu, H. E. Hoekstra, and N. E.
519 Pierce (2018). The genetic basis of a social polymorphism in halictid bees. *Nature*
520 *Communications* 9(1), 4338.

521 Kraft, N. J., O. Godoy, and J. M. Levine (2015). Plant functional traits and the mul-
522 tidimensional nature of species coexistence. *Proceedings of the National Academy*
523 *of Sciences* 112(3), 797–802.

524 Lee, I. P. A., O. T. Eldakar, J. P. Gogarten, and C. P. Andam (2022). Bacterial co-
525 operation through horizontal gene transfer. *Trends in Ecology & Evolution* 37(3),
526 223–232.

527 Lewontin, R. C. (1961). Evolution and the theory of games. *Journal of Theoretical*
528 *Biology* 1(3), 382–403.

529 Lieberman, E., C. Hauert, and M. A. Nowak (2005). Evolutionary dynamics on
530 graphs. *Nature* 433(7023), 312–316.

531 Lin, H., D. Wang, Q. Wang, J. Mao, Y. Bai, and J. Qu (2024). Interspecific compe-
532 tition prevents the proliferation of social cheaters in an unstructured environment.
533 *The ISME Journal* 18(1), wrad038.

534 May, R. M. (1987). More evolution of cooperation. *Nature* 327(6117), 15–17.

535 Michel-Mata, S., M. Kawakatsu, J. Sartini, T. A. Kessinger, J. B. Plotkin, and C. E.
536 Tarnita (2024). The evolution of private reputations in information-abundant
537 landscapes. *Nature* 634(8035), 883–889.

538 Motro, U. (1991). Co-operation and defection: playing the field and the ESS. *Journal*
539 *of Theoretical Biology* 151(2), 145–154.

540 Nowak, M. A. (2006). Five rules for the evolution of cooperation. *science* 314(5805),
541 1560–1563.

542 Nowak, M. A. and R. M. May (1992). Evolutionary games and spatial chaos. *na-*
543 *ture* 359(6398), 826–829.

544 Nowak, M. A. and K. Sigmund (1998). Evolution of indirect reciprocity by image
545 scoring. *Nature* 393(6685), 573–577.

546 Nowak, M. A. and K. Sigmund (2005). Evolution of indirect reciprocity. *Na-*
547 *ture* 437(7063), 1291–1298.

548 Nowak, M. A., C. E. Tarnita, and T. Antal (2010). Evolutionary dynamics in struc-
549 tured populations. *Philosophical Transactions of the Royal Society B: Biological*
550 *Sciences* 365(1537), 19.

551 Page, K. M. and M. A. Nowak (2002). Unifying evolutionary dynamics. *Journal of*
552 *theoretical biology* 219(1), 93–98.

553 Palmer, J. D. and K. R. Foster (2022). Bacterial species rarely work together. *Sci-*
554 *ence* 376(6593), 581–582.

555 Park, S. W., S. Cobey, C. J. E. Metcalf, J. M. Levine, and B. T. Grenfell (2024).
556 Predicting pathogen mutual invasibility and co-circulation. *Science* 386(6718),
557 175–179.

558 Park, S. W., J. Levine, and B. Grenfell (2026). Unifying coexistence theory for
559 ecological communities and pathogen strain competition. *bioRxiv*, 2026–05.

560 Pollak, S., M. Gralka, Y. Sato, J. Schwartzman, L. Lu, and O. X. Cordero (2021).
561 Public good exploitation in natural bacterioplankton communities. *Science Ad-*
562 *vances* 7(31), eabi4717.

563 Rakoff-Nahoum, S., K. R. Foster, and L. E. Comstock (2016). The evolution of
564 cooperation within the gut microbiota. *Nature* 533(7602), 255–259.

565 Schuster, P. and K. Sigmund (1983). Replicator dynamics. *Journal of theoretical*
566 *biology* 100(3), 533–538.

567 Sieben, A. J., J. R. Mihaljevic, and L. G. Shoemaker (2022). Quantifying mechanisms
568 of coexistence in disease ecology. *Ecology* 103(12), e3819.

569 Slobodkin, L. B. (1964). The strategy of evolution. *American scientist* 52(3), 342–
570 357.

571 Smith, J. M. and G. R. Price (1973). The logic of animal conflict. *Nature* 246(5427),
572 15–18.

573 Souza, M. O., J. M. Pacheco, and F. C. Santos (2009). Evolution of cooperation
574 under N-person snowdrift games. *Journal of Theoretical Biology* 260(4), 581–588.

575 Steidinger, B. S. and J. D. Bever (2014). The coexistence of hosts with different
576 abilities to discriminate against cheater partners: an evolutionary game-theory
577 approach. *The American Naturalist* 183(6), 762–770.

578 Szabó, G. and G. Fath (2007). Evolutionary games on graphs. *Physics reports* 446(4-
579 6), 97–216.

580 Tarnita, C. E. and A. Traulsen (2025). Reconciling ecology and evolutionary game
581 theory or “When not to think cooperation”. *Proceedings of the National Academy*
582 *of Sciences* 122(14), e2413847122.

583 Taylor, P. D. and L. B. Jonker (1978). Evolutionary stable strategies and game
584 dynamics. *Mathematical biosciences* 40(1-2), 145–156.

585 Traulsen, A. and M. A. Nowak (2006). Evolution of cooperation by multilevel selec-
586 tion. *Proceedings of the National Academy of Sciences* 103(29), 10952–10955.

587 Trivers, R. L. (1971). The evolution of reciprocal altruism. *The Quarterly review of*
588 *biology* 46(1), 35–57.

589 Wang, G., Q. Su, L. Wang, and J. B. Plotkin (2025). Evolution of social behaviors
590 in noisy environments. *arXiv*.

591 Wedekind, C. and M. Milinski (2000). Cooperation through image scoring in humans.
592 *Science* 288(5467), 850–852.

593 West, S. A., G. A. Cooper, M. B. Ghoul, and A. S. Griffin (2021). Ten recent insights
594 for our understanding of cooperation. *Nature ecology & evolution* 5(4), 419–430.

595 West, S. A., A. S. Griffin, and A. Gardner (2007). Evolutionary explanations for
596 cooperation. *Current biology* 17(16), R661–R672.